Beyond Offline Accuracy: Quantifying the Deployment Cost of Real-Time HRV Stress Inference

***Abstract*—Accuracies reported for heart-rate-variability (HRV) stress classification are invariably obtained offline using full-length sessions. Such assessments do not demand that a prediction be ready by the time it describes. However, a wearable system has tighter requirements, including not fitting per-session models, not accessing future data, and making predictions in bounded latency. In this paper, we quantify the penalties of such restrictions using a four-class stress task on the WESAD dataset. We decompose the gap between offline and deployable methods into distinct factors. The largest single factor is not algorithmic but the choice of labelling scheme. The commonly used midpoint scheme relies on samples recorded after the labelled instant, inflating macro-F1 by $+0.071$ to $+0.084$ across four independent architectures. Shrinking the analysis window to 60 beats from 120 leads to a latency of approximately 45 seconds and does not harm the performance; in fact, macro-F1 improves by $+0.034$ to $+0.075$. The first three commonly believed enhancements have no benefit at all: per-user calibration, a dedicated causal architecture, and short-horizon prediction. The fourth enhancement does precisely the opposite. Initializing the baseline of a new user from the mean of the entire population works much better than using the subject's own session mean. Given these findings, we argue that the deployment configuration must be chosen based on causal safety instead of minor differences in performance that are overshadowed by the variance of the seeds. We thus report the operating point of the resulting system rather than its performance in the offline setting.**

***Index Terms*—heart rate variability, stress detection, real-time inference, causal feature extraction, wearable systems, deployment cost**

I. Introduction

Stress detection using HRV from wearables is most often assessed offline using a batch classification task. The dataset is segmented into windows, features are extracted from each window, models are cross-validated, and finally, a single accuracy metric is reported. Yet, this procedure does not involve any requirement on how to compute a prediction at a certain point in time based on available information at that very point in time.

Deployed systems have tighter constraints. First, beats arrive sequentially, and second, a model cannot be fit to a session that has not completed yet. Finally, a moment in time cannot be labeled using evidence collected after the considered moment. Each constraint limits the amount of information or processing possible that could be used in offline evaluation.

It is rarely quantified how big of a limitation this is. A systematic review of wearable stress detection literature ranked deployment considerations as the worst evaluated aspects of all considered assessment domains and noted that there was not much consideration about deployment in practice [1]. What is needed is a decomposition of how costly each constraint is.

This paper offers such a decomposition for one concrete system: a four-level stress classifier which drives an adaptive virtual reality meditation app. For this system, predictions have to be made continuously, and have to happen within a minute from the current user state.

The contributions of this paper are as follows. Firstly, there is a labeling scheme that gives an overly optimistic macro-F1 score of around 0.08 for all architectures tested. The phenomenon is not exclusive to our architecture alone. Second, we demonstrate that reducing the length of the analysis window improves accuracy instead of decreasing it, and reduces the latency by half. Thirdly, three negative findings are reported: per-user fine tuning, a specially designed causal framework, and short-term forecasting. Additionally, a reversal is reported wherein population-level seeding is significantly better than the non-deployable within session approach.

II. Related Work

Research in wearable stress recognition is characterized by offline assessment with benchmark corpora. WESAD [2] still stands as the leading benchmark corpus. The reported performance varies a great deal according to task granularity, validation methodology, and windowing, making cross-paper comparisons complicated. Attention-based hybrid models can be considered exemplary of the current state of research in the field [3]. Newer corpora introduce additional sensing modalities [4], however, with the same old methodologies for evaluation. Benchmark comparisons of HRV features and classifiers were presented as well [5]. Large-scale studies of everyday life stress phenotypes were published too [6]. Nevertheless, both types of studies are retrospective.

A related body of work considers the problem of physiological inference in the context of a stream. Here, out-of-lab operation is a design consideration rather than simply an evaluation one. Continuous systems that can handle modal artifacts in real-life conditions have been developed [7]. The criterion of time to detection has also been considered as part of the evaluation process in addition to accuracy when building wearable systems [8]. Physiological foundation models offer yet another approach. Some have been pretrained using field data to bridge the lab-field gap [9], [10]. Yet this is largely a matter of representation rather than the causality and latency

considered herein. The fundamental constraint is well-known. At any point in time t, any statistic employed should be calculable using samples collected up to t. What has not been explored before is the quantification of the effect of imposing such a constraint on a model trained without such a constraint.

Rank-ordering in stress labels has led to the use of rank-consistent heads in models [11]. Class imbalance in physiological datasets is solved with focal loss [12]. While both are applicable to the present model, neither considers the deployability aspect.

Temporal leakage is well-studied; endpoint labelling is common practice in certain time-series fields. Yet, the effect size of inflation caused by midpoint labelling for wearable-based stress classification, and its consistency across model families, have not been quantified, to the best of our knowledge.

III. Deployment Constraints

We enumerate the constraints on the target system because each one maps to a term in the measurements described below.

A. No future samples

Every feature contributing to the predictions for beat t must be computable from beats $1 \dots t$. This means whole-session normalization, centered rolling statistics, and models trained on the full session cannot be applied.

B. Latency bound

The application tailors its behavior according to the inferred user's state. A prediction about a moment far in the past has limited practical value. The tolerable lag therefore bounds the window size from above [8].

C. No offline calibration

A new user has no prior history. Therefore, the baseline must be initialized using information available at first contact and updated as new data arrive.

D. Fixed compute budget

Inference shares hardware with the rendering workload. Therefore, the system runs inference on a separate backend rather than on the headset. The model must also be small enough to run at the required prediction rate.

IV. Methodology

A. Data and labels

We employ WESAD corpus [2]. It offers electrocardiography of the chest with sampling rate of 700 Hz and skin temperature of the wrist of fifteen subjects. The participants underwent protocol that elicited stress with the help of modified Trier Social Stress Test [13]. It also contained rest, amusement, and guided meditation.

Inter-beat intervals are computed through R-peak detection. Suspicious intervals and sudden beat-to-beat differences are discarded, and the discarded data points are recovered via interpolation. Descriptors of time and frequency domains are calculated according to the definitions [14]. The stress condition of the protocol is partitioned into three levels of intensity based on the local short-term variability, resulting in four ordinal levels.

B. Windowing and the labelling scheme

Windows consist of 60 successive heartbeats and shift by 5 beats. The critical question here is what sample to choose for the window labelling. There are two possible schemes:

- **Midpoint**: $y = \ell[s + W/2]$, the label at the window's centre.
- **Endpoint**: $y = \ell[e - 1]$, the label at the window's final beat.

The distinction concerns the *position* at which an existing annotation is read, not how that annotation is constructed. Labels are static offline products, and are therefore not computed in the deployed system. Thus, no causality is required in their construction. It is important to note that the constraint can also be overly broadly applied. Changing our tier statistic to recompute the intensity tiers from a past-only 30 beat window, rather than the centered window utilized throughout, increased the macro-F1 score by about 0.056. No other aspect of the deployed system was altered. The past-only tier statistic is mathematically closer to the variance features of the window itself, and thus the coupling of the target and the input increases. This represents a second distinct possibility for how windowed evaluation can lead to overly optimistic results. We report it so that future work can avoid the same issue.

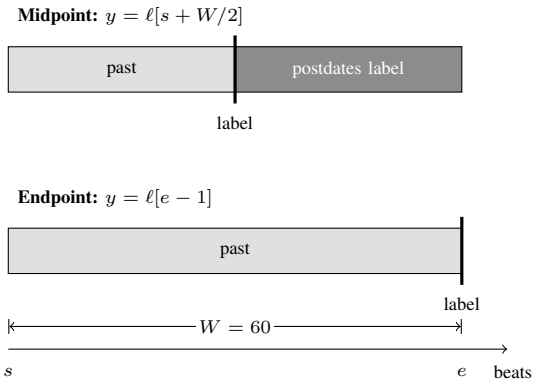


Fig. 1. Two ways of labelling a 60-beat window. Using the midpoint method, the label is determined at $s + W/2$. Consequently, the second half of the window (in dark) represents beats that have been captured after the label. A live system does not have access to such information. In the case of an endpoint method, the label is determined at the end of the window.

According to the mid-point convention, the model estimates a point when half of the inputs have arrived after the point in question (Fig. 1). For $W = 60$ beats, it means the estimation of 22 seconds of data after the moment labeled. This method is applicable for retrospective calculations, but it is not possible to use it for online processing.

We thus employ the end-point convention everywhere. The time-of-day features are also referenced to the same point in time. Consequently, none of the feature components has a timestamp beyond the estimated point.

C. Causal baseline estimation

The offline analysis works by fitting a rhythm model to the entire session and estimating the deviation from it. This needs the whole recording and can not work in an online fashion. We use three exponentially weighted moving averages, whose half-lives are 60, 300, and 1800 beats. They correspond to fast autonomic changes, session-wise baseline, and slow baseline changes. Each baseline tracker is initialized from a population estimate upon first meeting the user, and re-estimated upon each beat. All normalization and variability estimations are done with past-only buffers as well. The reason why we treat this as a design constraint and not as a measured cost is that we have not compared whole session versus causal baseline estimation directly. Instead, section V-E measures the benefit of having a causal baseline altogether.

D. Models

Each window has a feature vector of 25 dimensions, which include time and frequency domain features of HRV, deviation statistics compared to the causal baseline, two baselines, and cyclic encoding of the time-of-day. This feature vector is classified via gradient boosting trees [15]. Additionally, a seven-channel sequence of normalised beats signals is classified using a neural model. We examine two architectures of the neural model. First is the bidirectional convolutional and recurrent network, which was used in the offline pipeline. The second is the MS-CGCA, which is a causal alternative of this model and it consists of dilated causal convolution layers, a unidirectional recurrent layer, and cross-attention, where the query vector is the baseline and time-of-day. Both models are trained with focal loss due to class imbalance [12].

E. Evaluation

All configurations are evaluated using leave-one-subject-out cross-validation. If any configuration uses some part of the held-out subject's data for calibration, all configurations in that particular comparison are evaluated based on the same reduced data set. This guarantees that the comparisons are directly comparable. The ensemble weights are chosen in each fold from the remaining subjects. They are never chosen based on the predictions being evaluated. The abstention threshold is likewise chosen independently in each fold. It is based on the training subjects' margins, but is applied to the held-out subject. In all cases, κ values have been used for quadratic weighting. In case the configurations are examined using five random seeds, we provide the mean and standard deviation of results. When comparing between different configurations, pairing is done between seeds, while pairing between baseline-seeding methods is done on 15 leave-one-subject-out folds. Since there are only five seeds being paired, the lowest two-sided Wilcoxon p-value obtained is 0.0625. Therefore, seed level comparison cannot achieve statistical significance. The p-values are reported with effect sizes to indicate a non-significant difference due to low statistical power.

V. Results

A. The labelling convention

Table I reports the difference between the two conventions, holding everything else constant, for four independent configurations.

TABLE I
INFLATION FROM MIDPOINT LABELLING RELATIVE TO ENDPOINT LABELLING. FIVE SEEDS PER CONFIGURATION; IDENTICAL WINDOWS, FEATURES AND FOLDS. κ IS QUADRATICALLY WEIGHTED.

Configuration	**Δ macro-F1**	**$\Delta\kappa$**
Gradient-boosted trees alone	$+0.084$	$+0.075$
MS-CGCA two-way	$+0.071$	$+0.034$
Bidirectional two-way	$+0.076$	$+0.034$
MS-CGCA three-way	$+0.079$	$+0.042$

The phenomenon is present in all configurations and occurs to a similar extent in all of these different models. These include both tree ensembles and two distinct neural architectures. The phenomenon thus occurs due to the labelling scheme and not due to a particular model type. Moreover, at around 0.08 macro-F1, the phenomenon is larger than most of the differences in architecture described in the literature.

The practical consequences are significant. Midpoint-labelled accuracy can overstate endpoint performance by approximately this amount, even without implementation errors. We therefore suggest that any windowed physiological classification system disclose the labelling scheme.

B. Window Length

By lowering the length of the window to 60 beats from 120 beats, the prediction latency is decreased from 90 to 45 seconds. This change increases the accuracy of predictions in both models tested, by $+0.034$ in the case of MS-CGCA and $+0.075$ for the bidirectional model. Thus, latency and accuracy are not conflicting factors in this case. We see no need for using the longer window. Yet, we did not test windows shorter than 60 beats.

C. Configuration comparison

Table II compares four deployable configurations under endpoint labelling.

TABLE II
DEPLOYABLE CONFIGURATIONS, ENDPOINT LABELLING, FIVE SEEDS, IDENTICAL EVALUATION WINDOWS. κ IS QUADRATICALLY WEIGHTED.

Configuration	**Macro-F1**	**κ**
Bidirectional two-way	0.5993 ± 0.0120	0.7897 ± 0.0232
MS-CGCA three-way	0.5991 ± 0.0109	0.7773 ± 0.0225
MS-CGCA two-way	0.5925 ± 0.0129	0.7755 ± 0.0174
Trees alone	0.5703 ± 0.0040	0.7229 ± 0.0036

The top three methods cannot be differentiated from each other. In each case, $p \geq 0.625$, and their differences are less than their respective seed standard deviations. However, the tree ensemble alone does separate from the rest, lagging behind by 0.022 to 0.029 macro-F1 and $p = 0.0625$ in each comparison, which is the smallest possible value with five

paired seeds. So, sequence modelling adds value, but accuracy cannot show any clear preference for one of the sequence models.

D. Personalisation and architecture

Personalisation. Incorporating an extra ensemble member from personalisation of the population model through fine-tuning on a calibration slice from each held-out subject yields a macro-F1 change of $+0.0066$ (Wilcoxon test over five seeds, $p = 0.625$), indicating no detectable difference at this sample size rather than equivalence. Using the midpoint convention, the inclusion of this extra ensemble member seemed to yield an improvement of $+0.031$. But this did not persist under a change of the labelling convention. It requires per-subject calibration buffers, runtime fine-tuning, and a third artifact. The two-way path must exist anyway for uncalibrated users, so we do not deploy it.

Novel architecture. MS-CGCA was conceived for this use case. Causal padding and unidirectional recurrence guarantee that none of the hidden states rely on a future sample. Yet, it is not an improvement over the bidirectional model which it replaced (-0.0068 macro-F1; Wilcoxon test over five seeds, $p = 0.625$). There is still a need to use MS-CGCA for an important reason. The bidirectional model is admissible in this case only because the label is set at the window's end position. In case the windowing strategy is changed, it can rely on future samples without producing an error. MS-CGCA cannot do that by definition. When two architectures achieve comparable accuracy, we select the one whose correctness is not reliant on maintaining an invariant somewhere else in the system.

E. Cold-start seeding

The new user lacks any historical data, hence the EWMA trackers should be seeded using available knowledge. Table III provides a comparison of four initialisation approaches in case of endpoint labelling.

TABLE III
COLD-START SEEDING OF THE CAUSAL BASELINE, ENDPOINT LABELLING, LEAVE-ONE-SUBJECT-OUT, GRADIENT-BOOSTED COMPONENT ALONE. MACRO-F1 IS POOLED OVER ALL EVALUATION WINDOWS; PAIRED COMPARISONS IN THE TEXT ARE COMPUTED PER SUBJECT ACROSS THE FIFTEEN FOLDS AND THEREFORE DIFFER SLIGHTLY FROM DIFFERENCES OF THE POOLED VALUES. κ IS QUADRATICALLY WEIGHTED.

Seeding strategy	Macro-F1	κ
No causal baseline	0.496	0.615
Population mean (deployed)	0.565	0.737
Donor-cluster match	0.563	0.731
Own session mean (not deployable)	0.552	0.730

There are two important conclusions from this study. First, using the causal baseline improves macro-F1 by $+0.068$ compared with using HRV feature values directly. Second, one might reasonably expect the subject's own session average to be a better choice for seeding personalised predictions. However, the population mean performs better by 0.0186 macro-F1 (own-session minus population $= -0.0186$; Wilcoxon test over fifteen subjects, $p = 0.0054$, $d = -0.71$; the population mean performs better for 11 out of 15 subjects).

Matching on clusters of donors provides no detectable benefit. In this scheme, a new user is seeded from the training subject with the closest mean RR over the opening beats. It achieves a margin of $+0.0077$ relative to the population mean (Wilcoxon test over 15 subjects, $p = 0.934$, $d = 0.20$). Under midpoint labelling, however, the same comparison rejected donor matching with $p = 0.0026$. Thus, this result is sensitive to the labelling convention. We therefore use the population mean because it is simpler.

These two findings are consistent when the time-scales are considered. The fast and medium trackers have half-lives of 60 and 300 beats, respectively, and therefore converge within several hundred beats, effectively forgetting their seed. Changing only the initial condition is consequently not expected to have a lasting effect. In contrast, the slow tracker has a half-life of 1800 beats, which is comparable to the total recording length. It therefore preserves its seed throughout the recording, and its level becomes part of the feature vector. When seeded from the population mean, the tracker captures the difference between the subject's level and the population. When seeded from the same subject's mean, that difference disappears.

We offer this as a conjecture rather than a measured mechanism. It predicts that the effect would diminish if the slow tracker's half-life were shortened or its level were removed from the feature vector, neither of which was tested here.

F. Confidence-gated band selection

If the difference between the two most likely classes is smaller than a certain threshold, a combination band is generated, rather than enforcing a label. Table IV presents the evaluation of different coverages. Thresholds are computed independently for each fold based on margins from training subjects. This way, no testing window influences its threshold. Consequently, the achieved coverage deviates slightly from the desired coverage.

TABLE IV
CONFIDENCE GATING. MACRO-F1 AND SEVERE-ERROR RATE ARE COMPUTED ON *answered* WINDOWS ONLY, SO BOTH IMPROVE MECHANICALLY AS COVERAGE FALLS. $n = 12{,}026$ WINDOWS AT FULL COVERAGE.

Coverage	Macro-F1	ΔF1	Severe	ΔSev.
1.000	0.5647	—	0.0674	—
0.950	0.5748	+0.0100	0.0606	−0.0068
0.899	0.5804	+0.0157	0.0522	−0.0151
0.849	0.5868	+0.0221	0.0435	−0.0239
0.799	0.5899	+0.0251	0.0337	−0.0336
0.750	0.5832	+0.0184	0.0236	−0.0437
0.699	0.5520	−0.0128	0.0121	−0.0552

The maximum of Macro-F1 on answered windows comes around 80% coverage, where it is better than full coverage by +0.0251. In addition, the severe-error rate is reduced from 0.0674 to 0.0337. The improvement in macro-F1 is small relative to the seed variance reported in Table II. However, the

reduction in the error rate is more practically meaningful. Beyond about 70% coverage, the improvement over full coverage becomes negative and then drops sharply. When coverage is less than 70%, macro-F1 deteriorates to 0.359 at 60% coverage and 0.250 at 50% coverage. This occurs because abstention leaves the rare classes empty and destabilises per-class F1.

Among the least confident 20% of predictions, 66.9% of errors happen between two consecutive classes ($n = 1{,}469$). These are the kinds of errors that will be absorbed by the merged band.

G. Short-term forecasting

An ability to forecast an increase in stress could help the consuming application take measures before the user encounters such stress. We, therefore, seek to establish if the model is able to predict the value of the stress at the H-step horizon ahead of time. We compare the model with a persistence benchmark.

TABLE V
FORECASTING VERSUS PERSISTENCE, ENDPOINT LABELLING. PERSISTENCE WINS AT EVERY HORIZON TESTED.

Horizon	Model F1	Persistence F1	Δ
$\approx$30 s	0.466	0.543	-0.077
$\approx$60 s	0.453	0.499	-0.046
$\approx$120 s	0.420	0.453	-0.033

This model is inferior to persistence at all horizons. In this particular data set, at these particular horizons, there is nothing to forecast.

H. Deployed operating point

The deployed configuration is a two-member ensemble of gradient-boosted trees and the causal network, using 60-beat windows with endpoint labelling. On the exported artifact, it achieves a macro-F1 of 0.5923 and a quadratic κ of 0.7525. This is the nested estimate, with the blend weight re-selected using the remaining folds. The pooled score of 0.6023 is optimistic by $+0.0100$ and is therefore not reported. This matches the five-seed estimate of Table II. The weight of the tree component depends on each export independently and is not constant. For all the exports tested, the weight of the tree component varied between 0.15 and 0.20. There are no significant performance changes and everything is within the seed range.

VI. DISCUSSION

The decomposition indicates that the offline-to-deployable gap is not one of algorithm design. Of all terms measured, the biggest single term was due to a labelling convention, costs nothing to correct, and no amount of increased model capacity can bridge this gap. For window length, where a tradeoff between latency and accuracy is typically assumed, none could be observed.

There were three modifications which would typically be described as contributions but that showed no effect beyond seed variance: special architecture design, personalization and short-term prediction. The third modification failed to beat a persistence baseline at any horizon. There was one modification whose effect was opposite to the effect of the first three. Personalized cold start seeding should have been at least equal to the population average and at best better. In reality, the population seed has proved to be much more efficient than a subject's own baseline which cannot be obtained at deployment time anyway. These are the results that matter for a deployment cost study.

Two results depend on the labelling convention rather than on the intervention being studied. Under midpoint labels, personalisation appeared to contribute $+0.031$, but this decreased to only $+0.0066$ under endpoint labels. Similarly, donor-cluster seeding was rejected at $p = 0.0026$ under midpoint labels, whereas no detectable difference was observed under endpoint labels. Thus, in both cases, the convention did not merely rescale the result; it changed the conclusion a practitioner would draw. This differs from the cold-start reversal above, which holds under the corrected convention and reflects a property of the baseline itself rather than how it was measured.

VII. LIMITATIONS

The measurements were done using a single corpus of fifteen subjects. Given this sample size, the paired tests have low power. Consequently, failure to find any difference does not mean that equivalence has been proved, as proving equivalence requires an equivalence test with a predefined margin. The cold start, confidence, and forecasting experiments were conducted using a single random seed. Differences smaller than the seed variation reported in Table II, approximately 0.012 macro-F1, should not be interpreted as meaningful performance differences.

The windows have a high degree of overlapping due to the choice of the step size. This results in the effective sample size being closer to the number of subjects than to the number of windows. The absolute values must therefore be considered with this limitation in mind. However, all the comparisons done in this work used the same windowing scheme.

The intensity split of the stress factor comes from a short term variance statistic included among the input features. As can be seen in Section IV-B, the degree of correlation depends on the windowing used for the statistic, and the absolute accuracy varies because of it. The same definition was used throughout all the results in this paper; therefore, the results are not impacted. But the absolute accuracies cannot be considered to be calibrated using an external ground truth or compared with papers having different definitions for their intensity levels.

Lastly, we have computed the overhead of deployability on a single target platform with one latency budget. The direction of the effects may generalise, but their magnitudes may not.

VIII. CONCLUSION

We quantified the cost of converting an offline HRV stress classifier to a real-time classifier and decomposed it according

to constraints. The most important one turned out to be the labelling convention which exaggerated the performance metric by about 0.08 macro-F1 across four architectures. It is an artefact of the measurement process rather than a deficiency of the model itself.

Reducing the analysis window led to higher performance while reducing the latency by a factor of two. Three techniques which were thought to boost performance actually failed to do so: user-level fine tuning, dedicated causal architecture, and short-term forecasting. The fourth experiment revealed the reverse: seeding of the population mean was superior to that of the subject's own full session mean, which could not have been available during deployment anyway. In two scenarios, the alteration in the labelling technique altered the practical conclusions, where the value added by personalization became practically negligible, and donor-cluster seeding moved from highly rejected to being no different.

It is recommended that real-time physiological systems include their labeling scheme along with the operating point of the pipeline used. When configurations cannot be differentiated statistically, the decision-making process must be done according to causally safe factors rather than those lying within the seed variance range.

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